



جامعة عفت
EFFAT UNIVERSITY

CS 3081 - Artificial Intelligence

Assignment#1 - Fall 2024

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1 Introduction

This system is a basic medical diagnosis tool designed to assist patients by diagnosing common symptoms (fever, cough, or headache). It operates as both a simple reflex agent and a rule-based agent that provides a diagnosis based on the symptoms inputted by the user. In addition to the diagnosis, the system offers medicine recommendations and helpful tips to alleviate symptoms and improve patient well-being. The system interacts with a semi-dynamic environment, where the patient's condition can change over time, but it doesn't track the progression. With its rule-based design, the system offers a straightforward and practical solution to basic health concerns, aiming to increase accessibility to quick diagnoses and advice.

2 Task Environment Analysis

The medical diagnosis system operates within a simplified and controlled environment where it interacts with a patient, diagnoses their condition, and provides medical advice.

External Environment

The external environment is the patient's health condition and the symptoms they present. The system relies on the input provided by the patient (or a healthcare provider) to assess the situation. The environment includes factors like:

- Symptoms such as fever, cough, or headache.
- The patient's overall health, which affects the diagnosis.
- The external medical environment, such as the availability of medicines and healthcare facilities, is not directly observable by the system in this simplified model.

PEAS (Performance, Environment, Actuators, Sensors) Framework

Performance Measure: The system's success is measured by its ability to correctly diagnose based on input symptoms and provide relevant recommendations for medicine and tips. The goal is to increase accuracy, patient satisfaction, and advice effectiveness.

Environment: The system operates in a fully observable medical environment. The patient presents symptoms (fever, cough, headache), and the system must diagnose based on these limited inputs. The environment is semi-dynamic because, although the system gives a static diagnosis based on current symptoms, the patient's health condition may worsen or evolve over time if untreated. However, this progression isn't tracked by the system.

Actuators: Actuators refer to the system's ability to respond and provide actionable advice. In this case, the actuators are the system's capacity to:

- Generate a diagnosis based on symptoms.
- Recommend medicines (like paracetamol for fever or cough syrup for a cough).
- Provide recovery tips (such as drinking fluids or resting in a quiet room).

Sensors: Sensors refer to the input mechanisms that the system uses to receive information. In this system, the sensors are the symptoms reported by the patient (fever, cough, or headache). These are entered as inputs into the system, which acts as a diagnostic sensor. The system only perceives what is input by the user; there are no advanced sensors like thermometers or medical tests.

2.1 Classification of the Environment

Properties of the Task Environment:

- **Observability:** The system operates in a fully observable medical environment because the patient presents symptoms (fever, cough, headache), and the system diagnoses based on these limited inputs.
- **Determinism:** The environment is deterministic because the same set of symptoms will always produce the same diagnosis, medicine recommendations, and tips. The system follows predefined rules to provide consistent outputs.
- **Episodicity:** The environment is episodic. Because each diagnosis is treated as a standalone case. The system does not track or use past inputs (e.g., previous symptoms) to influence future diagnoses. Each interaction is independent.
- **Semi-Dynamic:** The environment is semi-dynamic. The system provides a static diagnosis based on the current symptoms, but the patient's health condition may evolve or worsen over time if untreated. However, the system itself does not track this progression or adapt to it.
- **Discrete:** The environment is discrete because there are a limited number of symptoms (fever, cough, headache) and a corresponding limited set of possible diagnoses, medicine recommendations, and tips.
- **Known Environment:** The environment is known to the agent. A known environment means that the agent (or its designer) has full knowledge of how the environment operates and the possible outcomes of its actions. In the case of this medical diagnosis system:
 - The agent is designed with a complete understanding of the symptoms it will encounter (fever, cough, headache).
 - It knows exactly what diagnosis, medicine, and tips to provide for each set of symptoms.
 - There is no uncertainty in how the symptoms will be processed because the system follows predefined rules.

2.2 Agent Design Implications

The medical diagnosis system is structured as a simple reflex agent, diagnosing based on immediate inputs—specifically, the symptoms provided by the user—and generating a corresponding diagnosis, medicine recommendation, and tips for recovery. Since the system operates in a deterministic and discrete environment, the decision-making process is rule-based, meaning it does not require complex reasoning, memory, or learning from past experiences. This design ensures that the agent is efficient for quick, repetitive tasks, consistently providing the same outputs for identical inputs without the need for adaptive behavior or contextual understanding.

So the system is both a simple reflex agent and rule-based. As a simple reflex agent, it reacts to current inputs (symptoms) by immediately providing a diagnosis and advice. This reflexive behavior is driven by predefined rules, which map specific inputs directly to corresponding actions. These rules ensure that the system's behavior is consistent and predictable, as the same inputs always lead to the same outputs, regardless of prior interactions or evolving conditions.

3 Python Implementation

3.1 Code Implementation

```

1
2 # Basic medical diagnosis system for medicine and tips
3
4 class BasicMedicalDiagnosis:
5     def __init__(self):
6         self.diagnosis = ""
7         self.medicine = ""
8         self.tips = ""
9
10    # Function to diagnose based on symptoms and provide medicine and tips
11    def diagnose(self, symptoms):
12        if 'fever' in symptoms:
13            self.diagnosis = "Diagnosis: You might have a fever-related illness (e
14            .g., flu).\n"
15            self.medicine = "Recommended Medicine: Take paracetamol (Panadol) to
16            reduce fever.\n"
17            self.tips = "Tips:\n- Rest well.\n- Drink plenty of fluids.\n- Avoid
18            cold environments."
19        elif 'cough' in symptoms:
20            self.diagnosis = "Diagnosis: You might have a cough-related illness (e
21            .g., cold).\n"
22            self.medicine = "Recommended Medicine: Take a cough syrup like
23            Robitussin.\n"
24            self.tips = "Tips:\n- Drink warm water.\n- Use a humidifier.\n- Avoid
25            smoking."
26        elif 'headache' in symptoms:
27            self.diagnosis = "Diagnosis: You might have a headache-related illness
28            (e.g., migraine).\n"
29            self.medicine = "Recommended Medicine: Take ibuprofen or aspirin to
30            relieve pain.\n"
31            self.tips = "Tips:\n- Rest in a quiet, dark room.\n- Avoid screen time
32            .\n- Stay hydrated."
33        else:
34            self.diagnosis = "Diagnosis: Symptoms do not match fever, cough, or
35            headache.\n"
36            self.medicine = "No medicine recommendation available.\n"
37            self.tips = "Tips: Consider seeing a doctor for further evaluation."
38        return self.diagnosis, self.medicine, self.tips
39
40    # Test simulation
41    patient_symptoms_1 = ['fever'] # Sample input for a patient with fever
42    patient_symptoms_2 = ['cough'] # Sample input for a patient with cough
43    patient_symptoms_3 = ['headache'] # Sample input for a patient with headache
44
45    # Creating an instance of the diagnosis system
46    medical_agent = BasicMedicalDiagnosis()
47
48    # Diagnosing based on the symptoms and getting medicine and tips
49    diagnosis_1, medicine_1, tips_1 = medical_agent.diagnose(patient_symptoms_1)
50    diagnosis_2, medicine_2, tips_2 = medical_agent.diagnose(patient_symptoms_2)
51    diagnosis_3, medicine_3, tips_3 = medical_agent.diagnose(patient_symptoms_3)
52
53    # Output the diagnosis, medicine, and tips for each patient in multiple lines

```

```

44 print(f"Patient 1 Symptoms: {patient_symptoms_1}\n{diagnosis_1}{medicine_1}{tips_1
    }")
45 print(f"Patient 2 Symptoms: {patient_symptoms_2}\n{diagnosis_2}{medicine_2}{tips_2
    }")
46 print(f"Patient 3 Symptoms: {patient_symptoms_3}\n{diagnosis_3}{medicine_3}{tips_3
    }")

```

Listing 1: Basic Medical Diagnosis System

Brief Explanation: The system is a medical diagnosis tool that diagnoses common symptoms (fever, cough, headache) and provides actionable recommendations. The function `diagnose()` checks the input symptoms and generates three outputs: diagnosis, medicine recommendations, and tips. Each output is formatted using newline characters (`\n`) to display each sentence on a new line, ensuring clarity in the output.

3.2 Analysis of Results

Example Output for All Three Cases:

```

1
2 Patient 1 Symptoms: ['fever']
3 Diagnosis: You might have a fever-related illness (e.g., flu).
4 Recommended Medicine: Take paracetamol (Panadol) to reduce fever.
5 Tips:
6 - Rest well.
7 - Drink plenty of fluids.
8 - Avoid cold environments.
9
10 Patient 2 Symptoms: ['cough']
11 Diagnosis: You might have a cough-related illness (e.g., cold).
12 Recommended Medicine: Take a cough syrup like Robitussin.
13 Tips:
14 - Drink warm water.
15 - Use a humidifier.
16 - Avoid smoking.
17
18 Patient 3 Symptoms: ['headache']
19 Diagnosis: You might have a headache-related illness (e.g., migraine).
20 Recommended Medicine: Take ibuprofen or aspirin to relieve pain.
21 Tips:
22 - Rest in a quiet, dark room.
23 - Avoid screen time.
24 - Stay hydrated.

```

Listing 2: Example Outputs for All Three Cases

The system provides not only a diagnosis but also actionable advice in the form of medicines and practical tips for managing symptoms. This makes the system more useful and interactive, moving beyond a simple diagnosis tool to a basic treatment advisory system.

4 Conclusion

The designed medical diagnosis system functions as a simple reflex agent, relying on rule-based decision-making to diagnose patients based on the symptoms they present. The system operates in a fully observable and known environment, meaning it has complete and accurate knowledge of the inputs (symptoms) and how to respond to them. Each diagnosis is made without reference to past inputs or a need to adapt over

time, making the system efficient and reliable for its purpose. By providing not only diagnoses but also medicine recommendations and tips for feeling better, the system aims to offer immediate, actionable advice to patients with common health concerns. The simplicity of its design ensures consistency and ease of use, making it a practical tool for basic medical assistance.